

Key Equations — CT Projection-Domain Deep Learning

A reference sheet for the projection-domain (sinogram / raw-CT) work: the parallel-beam model, the exact fan-beam $\leftrightarrow$ parallel rebinning, the helical trajectory, the autocorrelation identity, the Gabor/curvelet feature basis, and the information ordering.

Notation

- $\mathbf{f}(\mathbf{x}, \mathbf{y})$ — attenuation field (the image); $\mathbf{F}$ — its 2-D Fourier transform.
 - θ, s — parallel-beam projection angle and detector offset.
 - β, γ — fan-beam **source angle** and **fan angle** of a ray.
 - u — flat-detector coordinate (projected to the isocenter plane).
 - D — source-to-isocenter distance.
 - h — helical pitch (table travel per full rotation).
 - (r_0, φ_0) — polar position of a point in the object.
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1. Parallel-beam Radon transform

$$p_\theta(s) = \int f(s \cos\theta - t \sin\theta, s \sin\theta + t \cos\theta) dt$$

A point mass at (r_0, φ_0) maps to a **sinusoid** in the sinogram:

$$s = r_0 \cos(\theta - \varphi_0)$$

2. Fourier-slice (projection-slice) theorem

$$(1\text{-D FT of } p_\theta)(\omega) = F(\omega \cos\theta, \omega \sin\theta)$$

3. Filtered back-projection (inversion)

$$f(x, y) = \int_0^\pi (p_\theta * H)(x \cos\theta + y \sin\theta) d\theta, H(\omega) = |\omega|$$

The ramp filter $|\omega|$ is the source of high-frequency noise amplification / ill-conditioning.

4. Fan-beam $\rightarrow$ parallel-beam rebinning (exact)

Equiangular (curved/third-generation detector), ray indexed by (β, γ) :

$$\theta = \beta + \gamma, s = D \sin \gamma$$

Equispaced (flat detector), ray indexed by (β, u) , with $\gamma = \arctan(u / D)$:

$$\theta = \beta + \arctan(u / D), s = u D / \sqrt{D^2 + u^2}$$

Both reduce to the same relation $s = D \sin \gamma, \theta = \beta + \gamma$ — the mapping is a **nonlinear** resampling (note the $\sin$), not an affine transform.

5. Helical (spiral) source trajectory

$$z(\beta) = (h / 2\pi) \beta \text{ (pitch } h \text{ per rotation)}$$

The source moves in z during rotation, so projections through a given axial plane are incomplete; a per-slice sinogram is an interpolated (partially reconstructed) object.

6. The sinogram encodes the image autocorrelation (Wiener-Khinchin + Fourier-slice)

Image autocorrelation and its spectrum:

$$R_f(\tau) = \int f(x) f(x + \tau) dx, (2\text{-D FT of } R_f) = |F|^2$$

The 1-D autocorrelation of one projection is a **radial slice of the 2-D power spectrum**:

$$r_\theta(u) = \int p_\theta(s) p_\theta(s + u) ds, (1\text{-D FT of } r_\theta)(\omega) = |F(\omega \cos\theta, \omega \sin\theta)|^2$$

Sweeping θ therefore fills $|F|^2$, whose inverse transform is R_f — equivalently:

$$r_\theta = \text{Radon}[R_f]$$

(Exact in parallel-beam; under fan-beam/helical geometry it holds only after the nonlinear rebinning of §4 and is treated as a conceptual guide.)

7. Gabor atom — and its curvelet generalization

$$\psi(x; f_0, \theta, \sigma, \varphi) = \exp(-|x|^2 / 2\sigma^2) \cdot \cos(2\pi f_0 (x \cdot n_\theta) + \varphi)$$

Jointly localized in space and frequency (uncertainty bound), selective in orientation θ and scale $\propto 1/f_0$. Under real divergent geometry the point-response is a curved ridge, so the matched basis generalizes from straight Gabor atoms toward **curvelet-type** elements (curvelet / curved-Gabor / shearlet) — still anisotropic, oriented, and scale-selective, but adapted to a curved locus.

8. Information ordering (data-processing inequality)

For a real scan, Y (diagnosis) $\rightarrow f \rightarrow p_{\text{raw}}$ (raw fan-beam helical data) $\rightarrow x$ (image) is a Markov chain, so:

$$I(Y; x) \leq I(Y; p_{\text{raw}})$$

For a **synthetic** sinogram $p_{\text{syn}} = R[x]$ (forward-projected from the image):

$$I(Y; p_{\text{syn}}) \leq I(Y; x) \leq I(Y; p_{\text{raw}})$$

The synthetic sinogram cannot exceed the image it came from — the honest bound behind the "representation and architecture, not fidelity" scope of the synthetic-data experiments.